

F.R.I.D.A.Y.

Flexible Recommendation Intelligence with Digital Assisted paYments

Abstract

With the rapid growth of digital technologies, businesses are increasingly shifting towards social media platforms to market and sell their products. However, many businesses still spend a significant amount of time and effort interacting with customers through phone calls, messages, and manual order management processes. These traditional methods can be time-consuming, inefficient, and difficult to scale as customer demand increases.

This project aims to address these challenges by developing an intelligent e-commerce assistant that streamlines the entire purchasing process within a single platform. The system enables customers to browse products, place orders, make payments, and track their orders through a simple and interactive conversational interface. By automating routine customer interactions, the platform reduces the workload on business owners while improving the overall customer experience.

The proposed solution integrates product discovery, order management, payment processing, and shipment tracking into a unified system. This eliminates the need for customers to switch between multiple applications or communication channels to complete a purchase. The platform provides a convenient, efficient, and user-friendly shopping experience while helping businesses manage customer interactions more effectively.

Overall, the project demonstrates how conversational AI and automation can be leveraged to simplify online commerce, improve operational efficiency, and support the growing trend of social media-driven business models.

Introduction

The rapid growth of the internet and social media platforms has transformed the way businesses interact with customers and sell their products. Today, a large number of small and medium-sized businesses rely on social media channels such as Instagram, WhatsApp, Facebook, and Telegram to showcase products and attract potential buyers. While these platforms provide an effective way to reach customers, managing inquiries, orders, payments, and delivery updates manually can become challenging as the business grows.

In many cases, business owners spend a significant amount of time responding to customer messages, answering product-related questions, confirming orders, sharing payment details, and providing order status updates. This manual process not only increases workload but also leads to delays, inconsistencies, and a less efficient customer experience. Customers

often need to communicate through multiple messages before completing a purchase, making the overall shopping process inconvenient.

To address these challenges, this project proposes an AI-powered conversational e-commerce assistant that automates and simplifies the entire purchasing workflow. The system enables users to browse products, obtain product information, place orders, complete payments, and track their orders through a single conversational interface. By integrating these functionalities into one platform, the system eliminates the need for repeated manual interactions and reduces operational overhead for businesses.

The proposed solution leverages modern conversational AI techniques, database integration, payment gateway services, and order management systems to create a seamless shopping experience. Customers can interact with the assistant naturally, while businesses can manage orders more efficiently and focus on growth rather than repetitive administrative tasks.

This project demonstrates how intelligent automation can improve customer engagement, streamline business operations, and support the growing trend of social media-driven commerce in the digital era.

Problem Statement

With the increasing adoption of social media platforms for product promotion and sales, businesses receive a large number of customer inquiries through messages and chats. Managing these interactions manually requires significant time and effort from business owners and customer support teams. Customers often need to wait for responses regarding product availability, pricing, order placement, payment procedures, and order status updates.

Traditional communication methods such as phone calls and manual messaging are inefficient, especially when handling multiple customers simultaneously. These processes can lead to delayed responses, human errors, inconsistent customer experiences, and increased operational workload. Additionally, customers are often required to switch between different platforms for browsing products, making payments, and tracking orders, resulting in a fragmented purchasing experience.

Therefore, there is a need for an automated and intelligent system that can handle customer interactions efficiently while providing a seamless end-to-end shopping experience. The system should enable customers to browse products, place orders, make payments, and track deliveries through a single platform, reducing manual effort and improving overall customer satisfaction.

Objectives

The main objectives of this project are:

1. To develop an intelligent conversational e-commerce assistant for handling customer interactions.
2. To provide a unified platform for product browsing, order placement, payment processing, and order tracking.
3. To reduce the manual effort required by businesses in managing customer inquiries and orders.
4. To automate routine customer support tasks and improve response times.
5. To simplify the purchasing process through a natural language conversational interface.
6. To integrate secure online payment functionality within the system.
7. To provide real-time order status and tracking information to customers.
8. To enhance customer experience by offering a seamless and convenient shopping journey.
9. To improve operational efficiency and scalability for businesses using social media and digital sales channels.
10. To demonstrate the practical application of conversational AI in modern e-commerce systems.

Current System Analysis

The rapid growth of the internet, smartphones, and digital payment systems has significantly transformed the e-commerce industry. Today, customers expect quick responses, seamless purchasing experiences, and instant access to information. At the same time, businesses are increasingly using social media platforms and messaging applications as sales channels to reach a larger audience.

Despite these advancements, many businesses still rely heavily on manual processes to manage customer interactions. Customers often contact businesses through messages, calls, or social media chats to inquire about products, place orders, confirm payments, and request delivery updates. Business owners or support staff must manually respond to each customer, verify order details, process payments, and provide tracking information.

As the number of customers increases, this approach becomes difficult to manage efficiently. Delayed responses, missed inquiries, repetitive communication, and human errors can negatively impact customer satisfaction. Furthermore, customers are frequently required to move between multiple platforms for browsing products, making payments, and tracking orders, which creates a fragmented and less convenient shopping experience.

The increasing scale of e-commerce activities demands a more efficient and automated approach. Modern consumers expect services to be available at all times, while businesses seek solutions that can reduce operational workload and improve service quality. The existing system lacks the level of automation necessary to meet these growing demands.

Proposed Methodology

The proposed system is designed as a multi-agent conversational e-commerce platform that automates customer interactions, product purchasing, payment processing, and order fulfillment. The architecture consists of specialized AI agents coordinated by a Supervisor Agent, enabling efficient handling of different customer requests.

1. User Query Processing

The workflow begins when a user sends a message to the system. The message is first received by the Supervisor Agent, which acts as the central coordinator of the platform. The supervisor analyzes the user's intent and determines which specialized agent should handle the request.

The Supervisor Agent performs the following tasks:

- Classifies user intent.
- Plans the execution flow.
- Routes requests to the appropriate agent.
- Maintains coordination between agents.

2. Sales Agent

When a user requests product information, recommendations, or browsing assistance, the query is forwarded to the Sales Agent.

The Sales Agent:

- Searches for relevant products.
- Retrieves product details from the product database.
- Sorts and ranks products based on popularity and relevance.
- Answers customer questions regarding products.
- Maintains short-term memory of recent interactions.

The agent stores important information such as selected products, user preferences, and recent conversation context. This enables more personalized and context-aware interactions.

After presenting products, the system performs a Human-in-the-Loop (HITL) confirmation step to determine whether the user intends to proceed with the purchase.

3. Buy Agent

Once the user confirms the purchase intention, the request is transferred to the Buy Agent.

The Buy Agent:

- Identifies the selected product.
- Retrieves available product variants.
- Determines pricing based on selected variants.
- Explains pricing details to the customer.
- Maintains product state information throughout the purchasing process.

This stage ensures that customers clearly understand product options and final pricing before proceeding to payment.

4. Payment Agent

After purchase confirmation, the Payment Agent manages all payment-related operations.

The Payment Agent performs the following functions:

- Generates payment links.
- Creates checkout sessions.
- Calculates the final payable amount.
- Verifies payment status.
- Confirms successful transactions.
- Handles refund-related operations when required.

The system integrates with the Razorpay payment gateway to ensure secure and reliable payment processing.

5. Payment Verification and Confirmation

When a customer completes the payment, a webhook-based confirmation mechanism receives payment status updates from the payment gateway.

The confirmation process:

- Validates successful payment completion.
- Updates the order status in the database.
- Triggers the order fulfillment workflow.
- Notifies the customer regarding payment success.

This automated mechanism eliminates manual payment verification and reduces processing delays.

6. Shipping Agent

After successful payment verification, the Shipping Agent initiates the fulfillment process.

The Shipping Agent is responsible for:

- Processing shipment requests.

- Managing order dispatch information.
- Generating tracking details.
- Providing delivery status updates to customers.

Customers can subsequently request tracking information through the conversational interface.

7. Information Agent

Apart from product purchases, customers may have questions regarding business policies, services, delivery information, or general inquiries.

The Information Agent:

- Retrieves business-related information.
- Answers frequently asked questions.
- Provides customer support information.
- Assists users with non-purchase-related queries.

8. Memory and Context Management

The system incorporates short-term memory to maintain conversation context throughout the user session.

Stored information includes:

- Recently viewed products.
- Product identifiers.
- User preferences such as size, color, and variants.
- Previous conversation context.
- Latest user intent.

This memory mechanism improves conversational continuity and creates a more natural user experience.

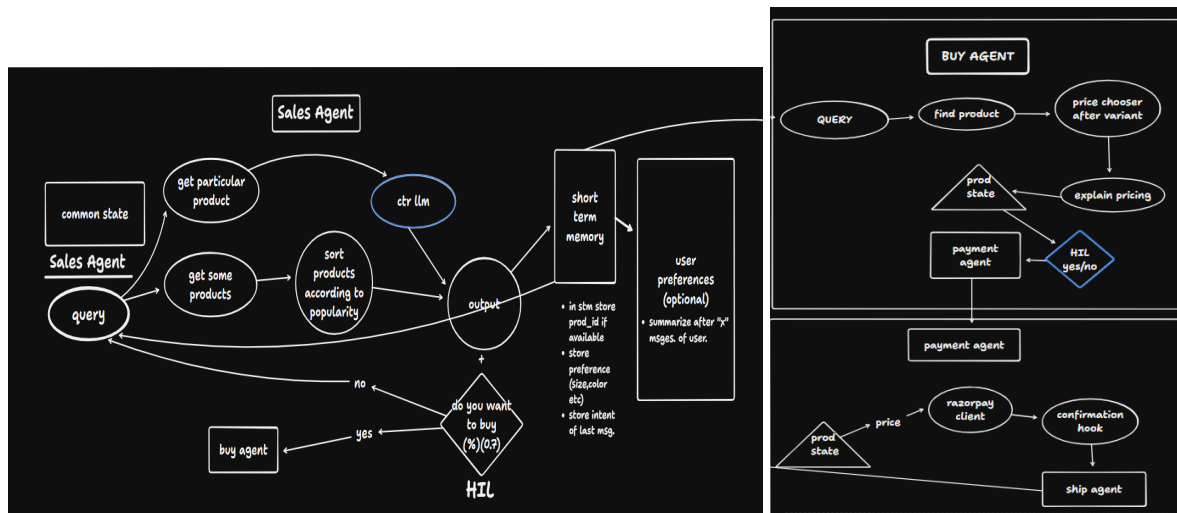
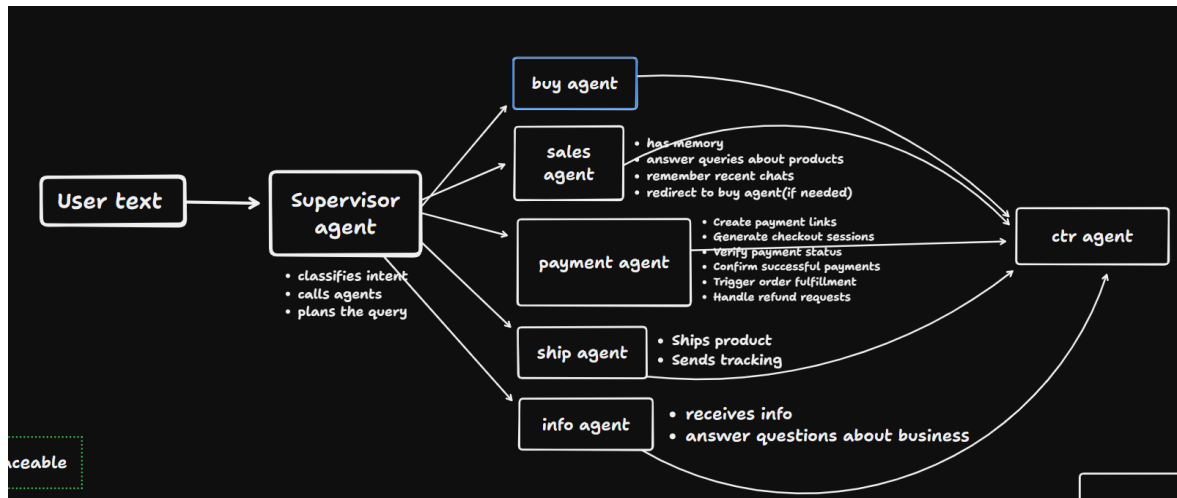
9. End-to-End Workflow

The complete workflow can be summarized as follows:

User Query → Supervisor Agent → Sales Agent → Purchase Confirmation (HITL) → Buy Agent → Payment Agent → Razorpay Payment Gateway → Payment Confirmation Webhook → Shipping Agent → Order Tracking and Delivery Updates

By distributing responsibilities among multiple specialized agents, the proposed methodology achieves scalability, modularity, automation, and an improved customer experience while reducing manual business operations.

System Architecture



Implementation

The proposed system was implemented using a multi-agent architecture in which each agent is responsible for a specific business function. The system processes user requests through a coordinated workflow managed by a central Supervisor Agent. This modular approach improves maintainability, scalability, and flexibility while enabling independent handling of different e-commerce operations.

1. Supervisor Agent Implementation

The Supervisor Agent acts as the entry point of the system. Every user message is first received by this module. The agent analyzes the user's request, determines the intent, and routes the query to the most appropriate specialized agent.

The Supervisor Agent is responsible for:

- Intent classification.
- Workflow planning.
- Agent selection.
- Inter-agent communication.

This implementation allows the system to dynamically choose the correct execution path without requiring hard-coded workflows for every scenario.

2. Sales Agent Implementation

The Sales Agent was implemented to handle product discovery and product-related conversations.

When a customer searches for a product, the Sales Agent:

- Retrieves product information from the database.
- Searches for products matching the user's request.
- Provides product descriptions and details.
- Answers customer questions regarding products.
- Stores product identifiers in short-term memory for future interactions.

The agent maintains conversational context, enabling users to continue discussions about products without repeatedly specifying product details.

3. Buy Agent Implementation

Once a customer decides to purchase a product, control is transferred to the Buy Agent.

The Buy Agent performs the following operations:

- Retrieves the selected product.
- Fetches available variants such as size, color, or configuration.
- Calculates the final price based on the selected variant.
- Explains pricing details to the customer.
- Stores product information required for payment processing.

This implementation ensures that customers receive accurate pricing information before proceeding with payment.

4. Payment Agent Implementation

The Payment Agent manages all payment-related operations within the system.

The module:

- Receives product and pricing information from the Buy Agent.
- Generates secure payment links.
- Creates checkout sessions.

- Monitors payment status.
- Handles transaction verification.

The payment process is isolated from other modules, improving security and simplifying payment management.

5. Razorpay Integration

The system integrates with the Razorpay payment gateway for transaction processing.

After receiving the final purchase details, the Payment Agent communicates with Razorpay to:

- Generate payment links.
- Process customer payments.
- Receive payment status information.

This integration enables secure and reliable online transactions while reducing manual payment handling.

6. Webhook-Based Payment Confirmation

To automate payment verification, a webhook mechanism was implemented.

When a customer completes a payment:

- Razorpay sends a payment confirmation event.
- The webhook receives the event.
- Payment status is validated.
- Order records are updated automatically.
- The shipping workflow is triggered.

This approach eliminates manual verification and reduces processing delays.

7. Shipping Agent Implementation

The Shipping Agent is activated after successful payment confirmation.

Its responsibilities include:

- Processing order fulfillment requests.
- Managing shipment information.
- Generating tracking details.
- Providing delivery status updates.

The agent ensures that customers can monitor the status of their orders after purchase.

8. Information Agent Implementation

The Information Agent was implemented to handle non-product-related customer inquiries.

The agent provides:

- Business information.
- Service details.
- Delivery policies.
- Frequently asked questions.

This separation prevents product-related agents from handling general support requests and improves overall system organization.

9. Memory Management

A short-term memory mechanism was implemented to maintain conversational continuity.

The memory module stores:

- Recent conversation history.
- Selected product identifiers.
- Current transaction context.
- Previously discussed product information.

This enables the system to understand follow-up queries and provide more natural interactions.

10. Overall Execution Flow

The implemented system follows the workflow shown below:

User Message → Supervisor Agent → Sales Agent → Buy Agent → Payment Agent → Razorpay Payment Gateway → Webhook Confirmation → Shipping Agent

For informational queries, the Supervisor Agent directly routes requests to the Information Agent. Throughout the interaction, short-term memory maintains context and supports seamless communication between modules.

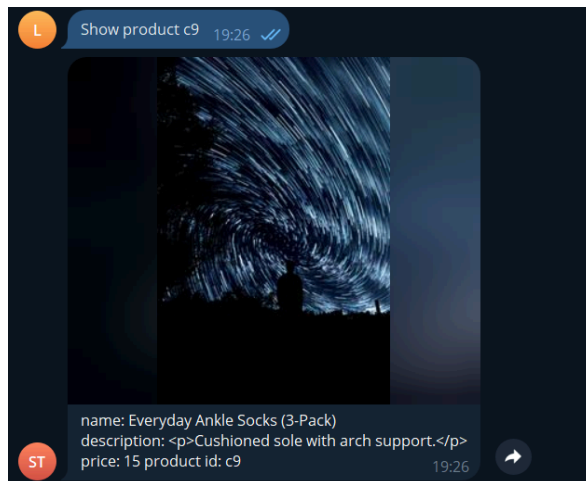
The modular implementation ensures that each agent performs a dedicated function while collaborating with other agents to deliver a complete end-to-end e-commerce experience.

Testing and Results

Test Case 1: Product Retrieval

Objective

To verify that the system can retrieve and display product information based on a user request.



Input

User Query: "Show product c-9"

Expected Result

The system should fetch the requested product from the database and display its details.

Actual Result

The Sales Agent successfully retrieved the product information and displayed:

- Product image
- Product name
- Product description
- Product price
- Product ID

Status

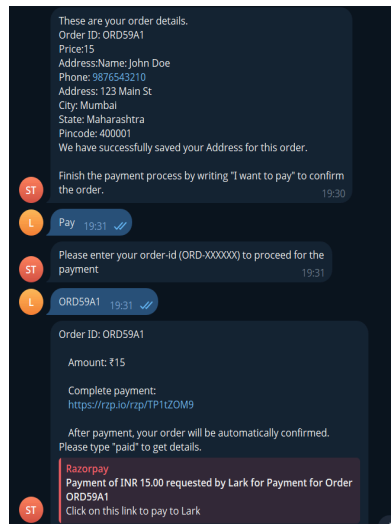
Passed

The result confirms that the product search and retrieval functionality operates correctly.

Test Case 2: Order Creation and Payment Link Generation

Objective

To verify that the system can create an order and generate a payment link.



Input

User selects a product and proceeds with the purchase process.

Expected Result

The system should:

- Generate a unique Order ID.
- Save customer details.
- Create a payment request.
- Generate a secure payment link.

Actual Result

The system successfully generated:

- Order ID: ORD59A1
- Order Amount: ₹15
- Customer Address Details
- Razorpay Payment Link

The customer was instructed to complete payment using the generated link.

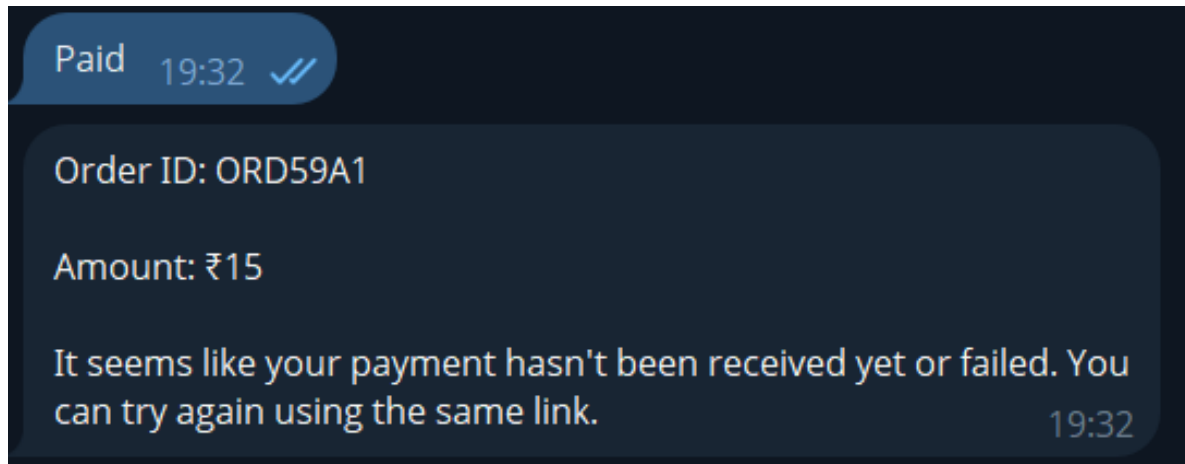
Status

Passed

Test Case 3: Failed Payment Verification

Objective

To verify the system's behavior when payment is not completed successfully.



Input

User enters "Paid" without a successful payment confirmation from the payment gateway.

Expected Result

The system should detect the incomplete payment and notify the customer accordingly.

Actual Result

The system displayed a message indicating that payment had not been received and advised the user to retry using the payment link.

Status

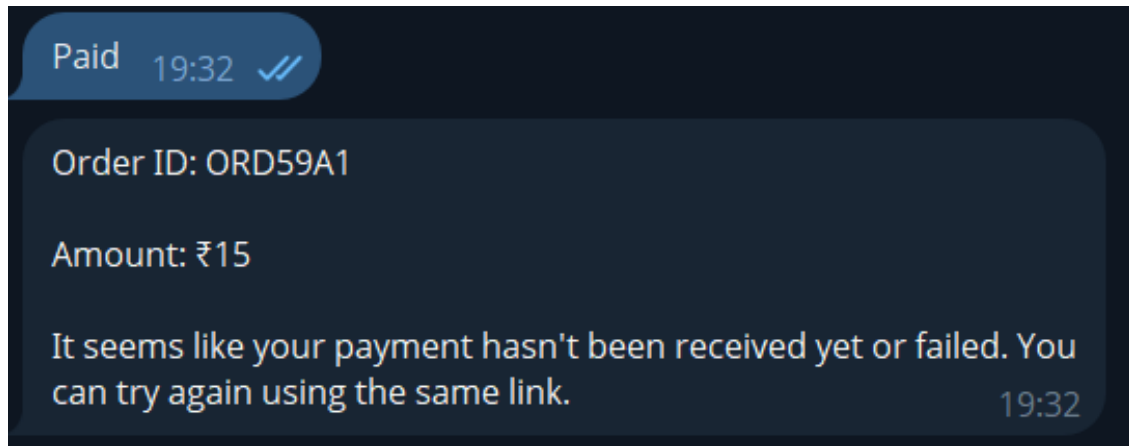
Passed

The webhook verification mechanism successfully prevented false payment confirmations.

Test Case 4: Successful Payment Verification

Objective

To verify successful payment detection and order confirmation.



Input

User completes payment and requests confirmation.

Expected Result

The system should validate the transaction and confirm the order.

Actual Result

The system successfully verified the payment and displayed:

- Order ID
- Payment Amount
- Payment Success Confirmation
- Order Dispatch Notification

The customer received confirmation that the order would be processed and shipped.

Status

Passed

The Payment Agent and webhook integration correctly verified the transaction and updated the order status.

Advantages

The proposed conversational e-commerce system offers several advantages over traditional manual sales and customer support processes.

1. **Reduced Manual Effort**
 - Automates customer interactions, reducing the need for constant human involvement in routine tasks.
2. **24/7 Availability**
 - Customers can browse products, place orders, and obtain information at any time without depending on business working hours.
3. **Faster Customer Response**
 - Provides instant responses to customer queries, improving user experience and reducing waiting times.
4. **Centralized Shopping Experience**
 - Enables product browsing, order placement, payment processing, and order tracking within a single conversational platform.
5. **Automated Payment Processing**
 - Generates payment links automatically and verifies transactions without manual intervention.
6. **Real-Time Order Updates**
 - Customers receive order and payment status updates promptly, improving transparency and trust.
7. **Improved Scalability**
 - The system can handle multiple customer interactions simultaneously, making it suitable for growing businesses.
8. **Modular Multi-Agent Architecture**
 - Individual agents handle specific tasks, making the system easier to maintain, upgrade, and expand.
9. **Enhanced Customer Experience**
 - Natural language interaction simplifies the purchasing process and makes the platform user-friendly.
10. **Reduced Human Errors**
 - Automation minimizes mistakes in order processing, payment verification, and customer communication.
11. **Efficient Order Management**
 - Orders, payments, and shipment information are processed systematically, improving operational efficiency.
12. **Business Growth Support**
 - Allows businesses to focus on product development and sales growth rather than repetitive customer service activities.

Limitations

Although the proposed system successfully automates several e-commerce operations, certain limitations still exist and provide opportunities for future improvement.

1. State Management Dependency

- The system relies on short-term memory to maintain the current workflow state. If critical state variables such as purchase flow or payment flow indicators are incorrectly modified or become inconsistent, the conversation may remain within a specific workflow instead of returning to the Supervisor Agent for proper intent re-evaluation.

2. Workflow Recovery Challenges

- Once a user enters a purchase or payment process, the system may have difficulty automatically recovering from unexpected conversation paths. More robust state management and fallback mechanisms are required to handle such scenarios effectively.

3. Prompt Engineering Limitations

- The performance of the agents depends heavily on prompt design. Ambiguous or unforeseen user inputs may occasionally lead to incorrect interpretations or suboptimal responses.

4. Dependency on External Services

- Payment processing depends on third-party services such as Razorpay. Service interruptions or API failures may temporarily affect transaction processing.

5. Scalability Considerations

- As the number of users increases, additional optimization and infrastructure scaling may be required to maintain system performance.

6. Incomplete Exception Handling

- Certain edge cases and uncommon user behaviors may not yet be fully covered, potentially leading to unexpected conversation flows.

7. Need for Advanced Routing Mechanisms

- Future versions should incorporate more intelligent routing and state-transition mechanisms to ensure that users can seamlessly switch between different tasks without becoming locked into a particular workflow.

Future Scope

The proposed conversational e-commerce system establishes a strong foundation for automated customer interactions. However, several enhancements can be incorporated in future versions to improve functionality, reliability, and customer satisfaction.

1. **Dedicated Support Agent**
 - A specialized Support Agent can be introduced to handle situations where other agents fail to provide satisfactory responses. This agent can assist users in resolving issues and improving the overall reliability of the system.
2. **Delivery and Shipping Assistance**
 - Future versions can provide detailed responses regarding shipment status, estimated delivery dates, courier information, and delivery-related concerns through automated tracking integration.
3. **Business Information Handling**
 - The system can be enhanced to answer a wider range of business-related queries, including store policies, return and refund policies, working hours, service information, and contact details.
4. **Human Support Escalation**
 - A human support request mechanism can be implemented to allow customers to connect with a support representative whenever automated assistance is insufficient or when manual intervention is required.
5. **Payment Failure Resolution**
 - Future enhancements can enable the system to automatically detect payment failures and guide customers through alternative payment options or troubleshooting procedures.
6. **Money Deducted but Payment Not Confirmed Cases**
 - The system can be extended to handle transaction disputes where money has been deducted from the customer's account but payment confirmation has not been received. Automated ticket generation and support workflows can help resolve such issues efficiently.
7. **Advanced Workflow Management**
 - Improved state management and agent-routing mechanisms can be implemented to prevent users from becoming stuck in specific workflows and ensure smooth transitions between different tasks.
8. **Enhanced Memory and Personalization**
 - Future versions can maintain long-term customer preferences, purchase history, and interaction patterns to provide personalized recommendations and improved customer experiences.
9. **Multi-Platform Deployment**
 - The system can be deployed across multiple communication channels such as WhatsApp, Telegram, Instagram, Facebook Messenger, and web applications to increase accessibility.
10. **Analytics and Business Insights**
 - Additional modules can provide sales analytics, customer behavior insights, popular product reports, and performance dashboards to assist business owners in decision-making.

User Instruction Manual

Introduction

The E-Commerce Assistant is a conversational shopping platform that allows users to browse products, place orders, and complete payments through a chat interface. Users can interact with the system using natural language commands and follow the instructions provided by the assistant.

Step 1: Browse Products

To view available products, send a product-related query to the assistant.

Examples:

- Show products
- Show available products
- Show product c-9
- I am looking for socks
- Show me a phone

The assistant will display relevant product information including:

- Product Image
- Product Name
- Product Description
- Product Price
- Product ID

Step 2: Select a Product

After reviewing the displayed products, choose the product you wish to purchase.

The recommended method is to use the Product ID provided by the assistant.

Examples:

- I want product c-9
- Buy product c-9
- Purchase c-9

The assistant will identify the selected product and begin the purchasing process.

Step 3: Provide Order Information

The assistant may request additional information required for order processing.

Typical details include:

- Customer Name
- Phone Number
- Delivery Address
- City
- State
- Postal Code

Provide accurate information to ensure successful order delivery.

Step 4: Review Order Details

The assistant will display an order summary containing:

- Order ID
- Product Information
- Total Amount
- Delivery Address

Review all details carefully before proceeding.

If the information is correct, follow the assistant's instructions to continue.

Step 5: Initiate Payment

When instructed, request payment processing.

Example:

- Pay
- I want to pay

The assistant will generate a payment request and provide an Order ID if required.

If prompted, enter the Order ID exactly as displayed.

Example:

- ORD59A1

Step 6: Complete Payment

The assistant will provide a secure payment link.

Instructions:

1. Open the payment link.
2. Complete the payment through the payment gateway.
3. Return to the chat after payment is completed.

Step 7: Confirm Payment Status

After completing payment, notify the assistant.

Example:

- Paid

The system will verify the transaction status.

Possible outcomes:

Successful Payment

The assistant will display a payment success message and confirm that the order has been processed.

Unsuccessful Payment

If payment has not been received or verification fails, the assistant will inform you and may ask you to try again using the same payment link.

Important Notes

- Always use the Product ID when selecting a product.
- Ensure that delivery details are entered correctly.
- Do not modify the Order ID provided by the system.
- Complete payment only through the payment link generated by the assistant.
- Wait for payment confirmation before assuming the order has been placed.
- Follow the instructions displayed by the assistant at each step of the purchasing process.

Purchase Workflow Summary

Browse Products → Select Product → Provide Delivery Details → Review Order Information
→ Request Payment → Complete Payment → Verify Payment → Order Confirmation